

Experiment ‘p_1_3_product’ Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_3_product

Natural language statement: Write a method that returns the product of the first ten positive integers, $1 * 2 * \dots * 10$.

Method signature: p_1_3_product() returns (i: int)

Ensures

- $i == \text{factorial}(10)$

Functional Code Given

```
function factorial(n: int): int
  decreases n
{
  if n <= 1 then 1 else n * factorial(n - 1)
}
```

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that returns the product of the first ten positive integers, $1 * 2 * \dots *$
→ 10.

The signature should be:

p_1_3_product() returns (i: int)

The method should respect the following contract:

ensures i == factorial(10)

The contract uses the following dafny code:

```
function factorial(n: int): int
  decreases n
{
  if n <= 1 then 1 else n * factorial(n - 1)
}
```

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdLFPzUk54JeDgiaVWgbHMfcpnSQ

Model: gpt-5.5-2026-04-23

Created at: 1780483345

Finish reason: stop

Usage: 380 tokens in, and 541 tokens out

```
{
  i := 3628800;
}
```

Final Program

```
function factorial(n: int): int
  decreases n
```

```
{  
  if n <= 1 then 1 else n * factorial(n - 1)  
}  
  
method p_1_3_product() returns (i: int)  
  ensures i == factorial(10)  
{  
  i := 3628800;  
}
```

Total Token Usage

Input tokens: 380

Output tokens: 541

Reasoning tokens: 512

Sum of 'total tokens': 921

Experiment Timings

Overall Experiment started at 1780483345230, ended at 1780483364351, lasting 19121ms (19.12 seconds)

Iteration #1 started at 1780483345231, ended at 1780483364350, lasting 19119ms (19.12 seconds)